POLITECNICO
MILANO 1863

Visual reconstruction of played music

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I. Problem Statement

The objective of this project is to analyze a video of a played piano keyboard in order to identify the pressed keys at each instant. Given a monocular RGB video, the task is decomposed into multiple stages, including keyboard localization, identifying individual keys, tracking fingertip positions, and determining which keys are pressed over time.

The system is required to operate on standard video without any specialized sensors. All detections must be expressed in the full image coordinate system to allow interaction between different processing stages.

The main challenge arises from the presence of hands, reflections, and background elements, which make direct analysis on the full image difficult. For this reason, the processing pipeline is structured into multiple stages, where intermediate representations are used to simplify subsequent steps.

II. Method

A. Overall Pipeline

The proposed pipeline processes the input video frame by frame. First, representative frames are selected from the video. Then, the piano keyboard is localized and isolated. White and black keys are detected on the keyboard region. Fingertips are detected on the full image. Finally, fingertip positions are associated with keys to identify pressed keys at each time instant.

B. Frame Selection

From each input video, a subset of frames is extracted. Frames with minimal hand occlusion are selected. These frames are used for keyboard and key detection.

C. Keyboard Localization

The piano keyboard is localized in the selected frames. The keyboard region is isolated from the background to simplify further processing.

D. Piano Key Detection

White and black piano keys are detected inside the keyboard region. White keys are detected first to define the keyboard structure. Black keys are detected relative to the white key layout.

E. Fingertip Detection

Fingertips are detected on the full image. Detection is performed independently from key detection. This ensures that fingertip positions remain consistent with the original image coordinates.

F. Key-Finger Association

Detected fingertip positions are compared with key locations. A fingertip is assigned to a key if it lies inside the key region. Temporal consistency is used to reduce false detections.

G. Pressed-Key Detection

A key is considered pressed when a fingertip remains inside its region for a short temporal window. This allows key presses to be detected over time.

III. Implementation

A. Pipeline Structure and Intermediate Outputs

The implementation follows a three-stage pipeline. Each stage produces intermediate results that are reused by later stages. All intermediate outputs are saved to disk to allow modular processing and easier debugging. Table I summarizes the main intermediate outputs produced by each stage of the pipeline.

B. Frame Extraction and Hand-Free Frame Selection

Input videos are processed frame by frame. Frames are sampled at a fixed rate. Among the sampled frames, a frame with minimal hand occlusion is selected manually. This frame is used for keyboard and key detection.

TABLE I
PIPELINE STAGES AND SAVED INTERMEDIATE OUTPUTS

Stage	Saved Outputs
Preparation	Hand-free frame, keyboard ROI, masked keyboard image, white and black key coordinates (JSON), key overlay image
Hand analysis	Tracked fingertip positions, finger landmarks, per-finger press information (CSV)
Integration	Fingertip-to-key mapping, pressed keys per frame, final visualization video

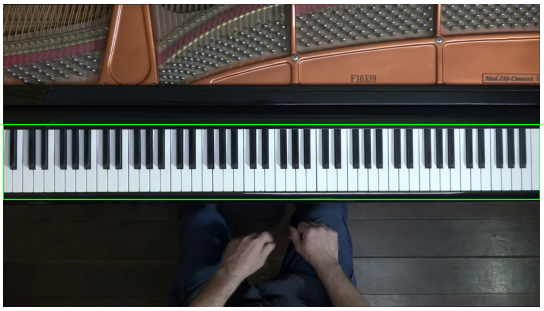


Fig. 1. Estimated keyboard region of interest (ROI) obtained from the hand-free frame.

C. Keyboard Region Localization and Masking

The piano keyboard region is localized using contour-based analysis. A bounding box is estimated to isolate the keyboard from the background. This bounding box defines the region of interest (ROI).

A binary mask is applied to the ROI. Only pixels belonging to the keyboard are preserved. Background regions are removed. This simplifies the subsequent key detection process.

D. White and Black Key Detection

Key detection is performed on the masked and cropped keyboard image. White keys are detected first. Vertical intensity transitions are analyzed to locate borders between adjacent white keys. The median distance between detected borders is used to estimate the average white key width.

Using this width, a regular grid of white key bounding boxes is generated. The height of white keys is estimated from the keyboard region dimensions and kept constant across the keyboard.

Black keys are detected relative to the white key layout. Only valid musical positions are considered. Detection is restricted to the upper portion of the white keys, where black keys are expected to appear.

All detected key coordinates are initially expressed in the cropped image reference frame. They are then mapped back to the full image reference frame using the known ROI position and scale. The final key locations are stored as JSON files.

E. Fingertip Detection and Tracking

Fingertips are detected on all video frames. Detection is performed on the full image. Temporal tracking and filtering are applied to reduce unstable detections. Tracked fingertip positions are saved for later processing.

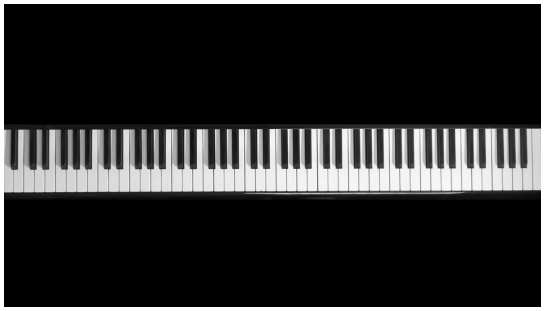


Fig. 2. Masked keyboard image after background removal.



Fig. 3. Detected white and black piano keys overlaid on the full image.

F. Key-Finger Mapping

For each frame, fingertip positions were compared with detected key bounding boxes. A fingertip was assigned to a key if it was located inside the corresponding key region. Multiple fingertip detections were handled independently.

G. Pressed-Key Detection

Key press events were detected over time. A key was considered pressed when a fingertip remained inside its region across consecutive frames. Short temporal smoothing was applied to reduce spurious detections.

IV. Experiments and Results

A. Dataset

The system was evaluated on multiple piano-playing videos. All videos were recorded using a static camera. Different video resolutions and quality levels were used. Both high-resolution and low-resolution videos were included. Some videos contained blur and compression artifacts. This was done to observe how the pipeline behaves under different visual conditions.

B. Qualitative Evaluation

Intermediate results were evaluated visually. The keyboard ROI, masked keyboard image, and key detection overlays were inspected. Detected white and black keys showed good alignment with the actual keyboard layout.

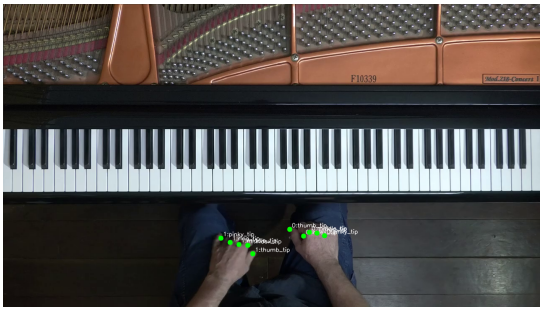


Fig. 4. Fingertip detection and key-finger association on a representative frame.

C. Integration Results

Detected key locations were combined with fingertip detection results. Since all outputs were expressed in full image coordinates, key–finger mapping was performed directly. Pressed keys were correctly identified in representative piano sequences.

D. Limitations

The method assumes a static camera and a fixed keyboard position. Strong reflections or severe occlusions may affect detection results. These cases were not addressed in this project.

V. Conclusion

A vision-based system for analyzing piano playing from video was implemented. The keyboard region was localized, piano keys were detected, and fingertip positions were associated with keys. Pressed keys were identified over time.

The modular structure of the pipeline allows each processing stage to operate in a consistent coordinate system. Future work may include handling camera motion and improving performance under challenging lighting.